

Limit cycle example: Van der Pol

$$\ddot{x} + \mu(x^2 - 1)\dot{x} + x = 0$$

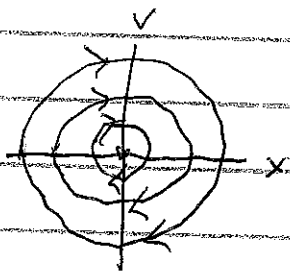
$\mu > 0$
(we will examine $\mu \gg 1$ case)

$\mu = 0$: $\ddot{x} + x = 0$

harmonic oscillator

$$x(t) = A \cos(t - \phi)$$

is general soln.



$$\begin{aligned}\dot{x} &= v \\ \dot{v} &= -x\end{aligned}$$

$$E = \frac{1}{2} v^2 + \frac{1}{2} x^2 = \text{constant if } \mu = 0$$

$\mu \neq 0$:

$$\begin{aligned}\dot{x} &= v \\ \dot{v} &= -\mu(x^2 - 1)v - x\end{aligned}$$

$v = x = 0$ is only
fixed pt.

$$\begin{aligned}\dot{E} &= v\dot{v} + x\dot{x} = v(\dot{v} + x) = v(-\mu(x^2 - 1)v) \\ &= -\mu v^2(x^2 - 1)\end{aligned}$$

$$\begin{aligned}\dot{E} &> 0 & \text{if } v \neq 0, & x^2 < 1 & \quad (E \text{ increases for small } x) \\ \dot{E} &< 0 & \text{if } v \neq 0, & x^2 > 1 & \quad (E \text{ decreases for large } x)\end{aligned}$$

Can prove it has a unique stable limit cycle (if $\mu > 0$). The proof is given in the textbook based on results for systems of the form $\ddot{x} + f(x)\dot{x} + g(x) = 0$

Liénard's eqn.

with certain conditions on $f(x)$ & $g(x)$.

Special coordinates for:

$$\ddot{x} + f(x)\dot{x} = -g(x)$$

let $F(x) = \int f(x) dx$

then $\frac{d}{dt} \left[\underbrace{\dot{x} + F(x)}_{=W} \right] = -g(x)$

~~$\ddot{x} = -g(x)$~~ $\Rightarrow \begin{cases} \dot{x} = W - F(x) \\ \dot{W} = -g(x) \end{cases}$

in our van der Pol example

$$f(x) = \mu(x^2 - 1)$$

$$g(x) = x$$

$$F(x) = \mu \left(\frac{x^3}{3} - x \right)$$

$$\Rightarrow \begin{cases} \dot{x} = W - \mu \left(\frac{x^3}{3} - x \right) \\ \dot{W} = -x \end{cases}$$

$$\Rightarrow \text{let } W = \mu y; y = \frac{1}{\mu} W$$

$$\begin{cases} \dot{x} = \mu \left(y - \frac{x^3}{3} + x \right) \\ \dot{y} = -\frac{1}{\mu} x \end{cases}$$

Vander Pol oscillator $\mu \gg 1$

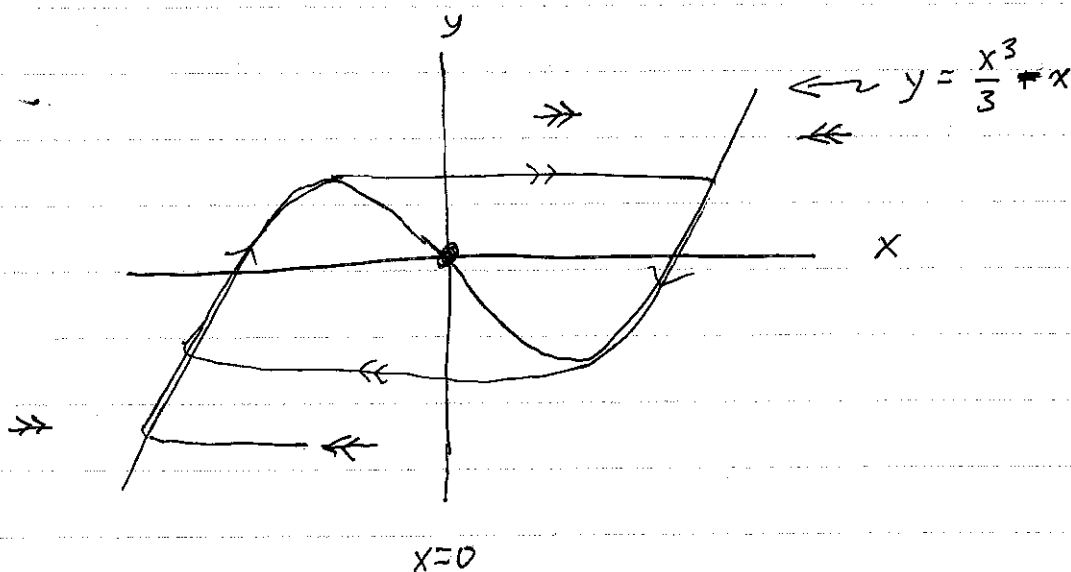
$$\ddot{x} + \mu(x^2 - 1)\dot{x} + x = 0$$

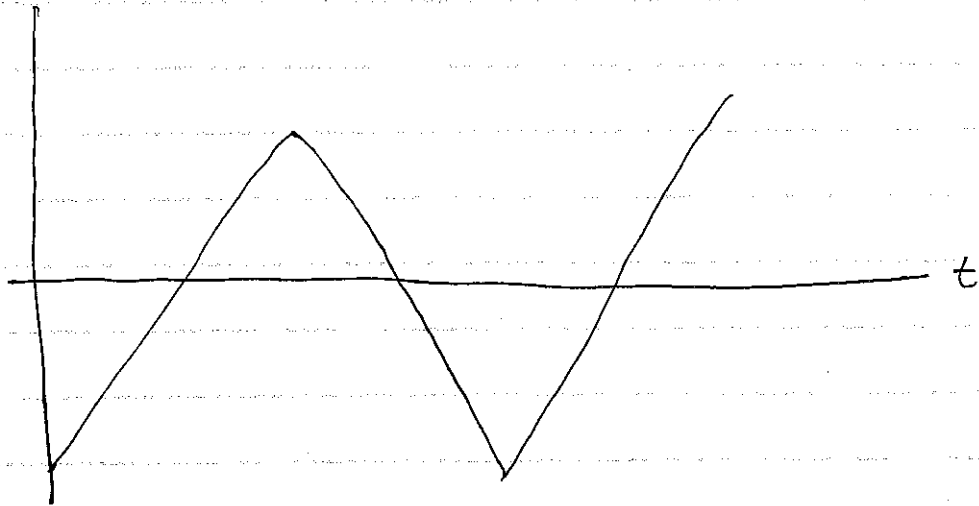
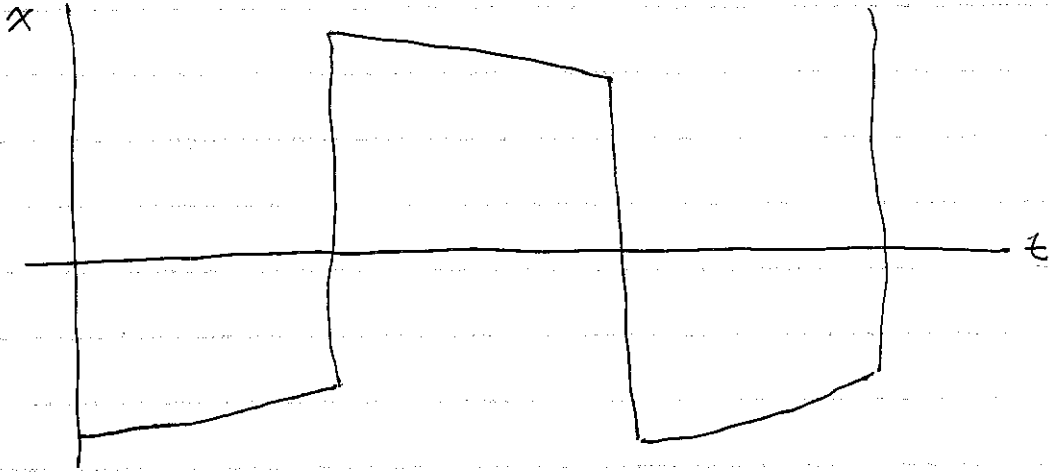
$$\ddot{x} + \mu(x^2 - 1)\dot{x} = -x$$

$$\frac{d}{dt} \left[\underbrace{\dot{x} + \mu \left(\frac{x^3}{3} - x \right)}_{\mu y} \right] = -x$$

$$\dot{x} = \mu \left[y - \frac{x^3}{3} + x \right] \quad \sim \mu \quad \text{Fast}$$

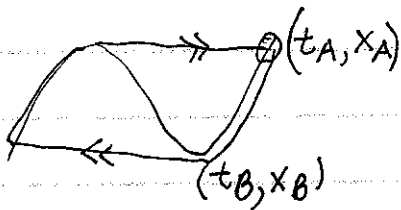
$$\dot{y} = -\frac{1}{\mu} x \quad \sim \frac{1}{\mu} \quad \text{slow}$$





estimating the period in the case that $\mu \gg 1$:

$$T \approx 2 T_{\text{slow}} = 2 \int_{t_A}^{t_B} dt$$



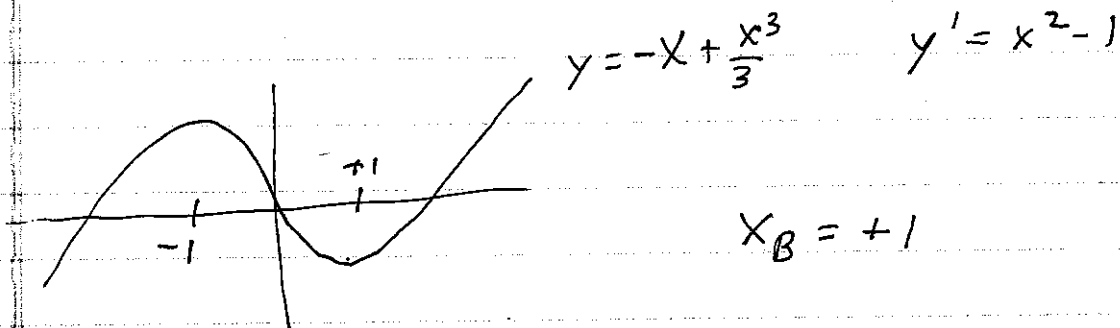
$$y \approx \frac{x^3}{3} - x \quad \text{on slow branch}$$

$$\Rightarrow \dot{y} = \dot{x}(x^2 - 1) = -\frac{1}{\mu} x$$

$$\int dx (x^2 - 1) \left(-\frac{\mu}{x}\right) = \int dt$$

$$T \approx -2\mu \int_{x_A}^{x_B} x - \frac{1}{x} dx$$

$$= 2\mu \int_{x_B}^{x_A} x - \frac{1}{x} dx$$



$$x_B = +1$$

$$y(-1) = -\frac{1}{3} + 1 = \frac{2}{3}$$

$$\frac{2}{3} = -x_A + \frac{x_A^3}{3} \Rightarrow x_A^3 - 3x_A - 2 = 0$$

$$(x_A + 1)(x_A^2 - x_A - 2) = 0$$

$$(x_A + 1)(x_A + 1)(x_A - 2) = 0$$

$$x_A = 2$$

$$T \approx 2\mu \int_1^2 x - \frac{1}{x} dx = 2\mu \left[\frac{x^2}{2} - \ln|x| \right]_1^2$$

$$= 2\mu \left(2 - \ln(2) - \frac{1}{2} \right)$$

$$T \approx \mu [3 - 2\ln(2)]$$

$$\approx 1.6\mu$$

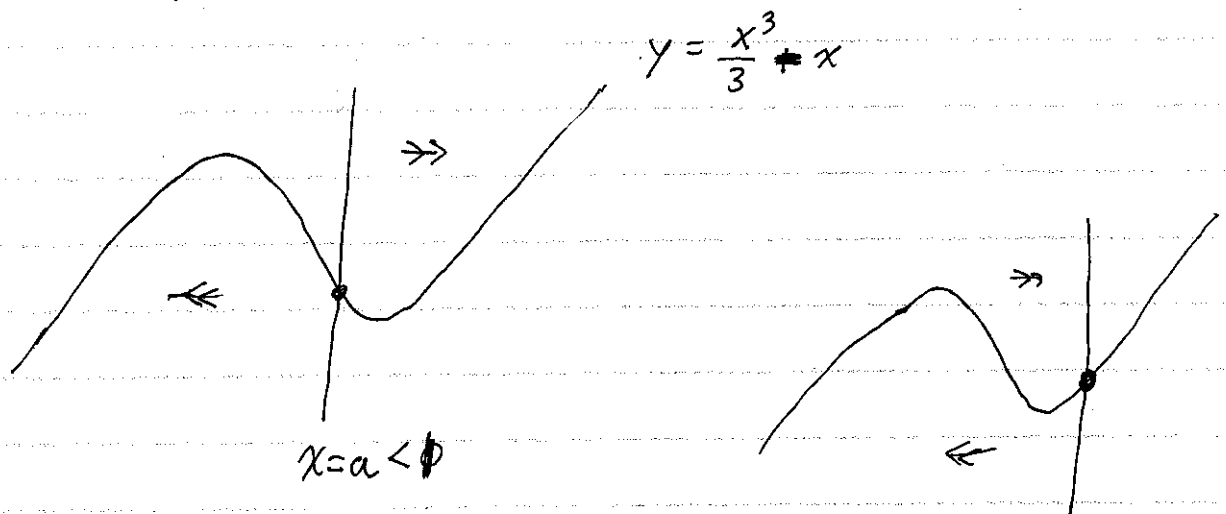
μ	T_{approx}	$T_{\text{numerical}}$
5	8.07	11.6
10	16.14	19.1
50	80.69	82.5
100	161.37	162.8

Biased van-der-Pol

$$\ddot{x} + \mu(x^2 - 1)\dot{x} + x = a \quad \Rightarrow \quad \frac{d}{dt} \left[\underbrace{\dot{x} + \mu \left(\frac{x^3}{3} - x \right)}_{\mu y} \right] = a - x$$

$$\Rightarrow \dot{x} = \mu \left[y - \frac{x^3}{3} + x \right]$$

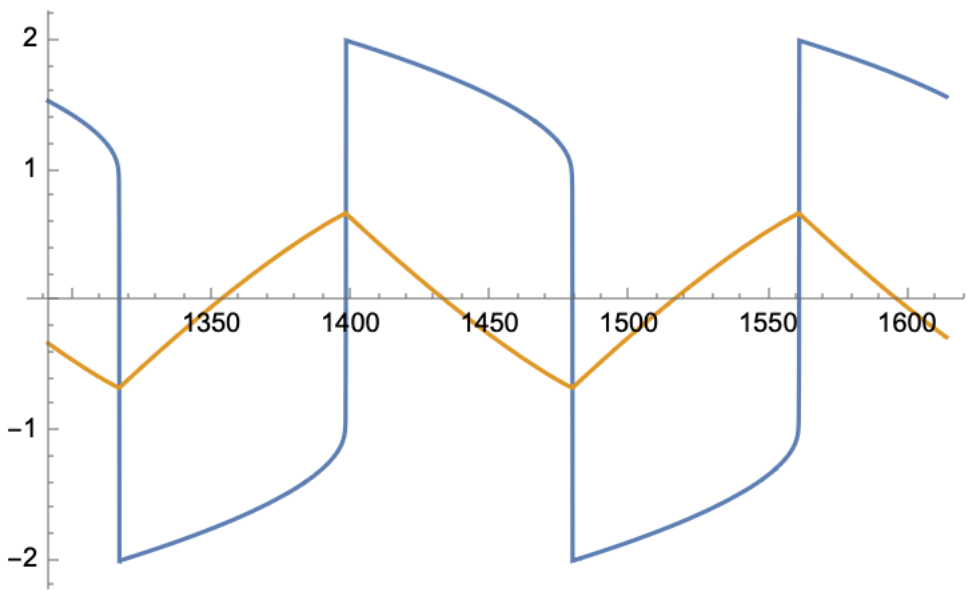
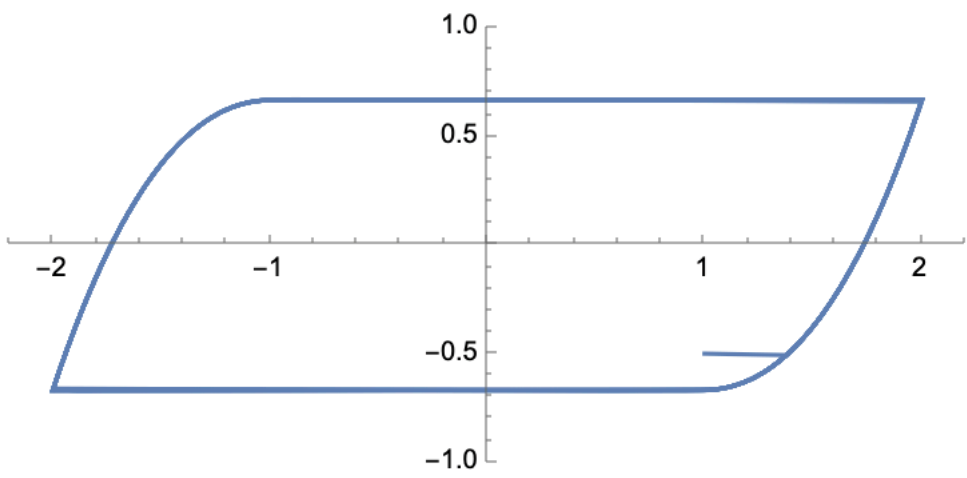
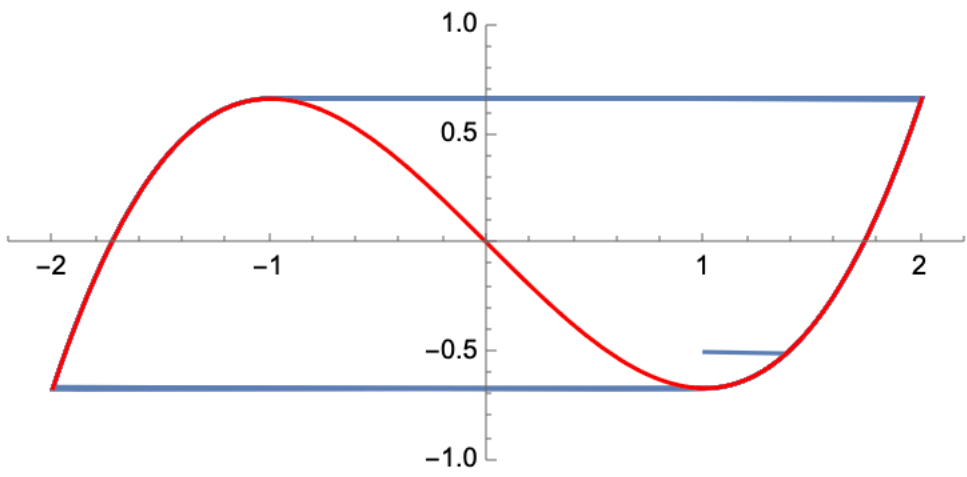
$$\dot{y} = \frac{1}{\mu} (a - x)$$



relaxation
oscillator

excitable system

```
mu = 100;  
Tapprox = N[(3 - 2 * Log[2]) * mu];  
s = NDSolve[{x'[t] == mu * (y[t] - x[t]^3/3 + x[t]), y'[t] == -(x[t])/mu, x[0] == 1.0, y[0] == -0.5}, {x, y}, {t, 10 * Tapprox}];  
  
Show[ParametricPlot[Evaluate[{x[t], y[t]} /. s], {t, 0, 10 * Tapprox}, PlotRange -> {{-2.2, 2.2}, {-1, 1}}],  
Plot[x^3/3 - x, {x, -2, 2}, PlotStyle -> Red]]  
ParametricPlot[Evaluate[{x[t], y[t]} /. s], {t, 0, 10 * Tapprox}, PlotRange -> {{-2.2, 2.2}, {-1, 1}}]  
Plot[Evaluate[{x[t], y[t]} /. s], {t, 8 * Tapprox, 10 * Tapprox}, PlotStyle -> Automatic]
```



```
Tmax = t /. FindMaximum[Evaluate[y[t] /. s], {t, 8 * Tapprox, 10 * Tapprox}][[2]]  
Tmin = t /. FindMinimum[Evaluate[y[t] /. s], {t, 8 * Tapprox, 10 * Tapprox}][[2]]  
period = 2 * Abs[Tmax - Tmin]  
period / mu  
  
1235.07  
  
1316.49|  
  
162.837  
  
1.62837
```